COSMICFORGE REAL ESTATE

Individual Property Owner Platform

Full Backend Implementation Specification

Implementation-ready engineering blueprint

Document control	Value
Specification version	1.0
Product source	Individual Property Owner Platform – Corporate Product Feature Document v5.0, 22 July 2026
Repository	CosmicforgeHealthnet/cosmicforge-realestate
Architecture baseline	11 database-owning microservices + API Gateway; no central identity service
Foundation baseline	F1–F5 certified at development/staging SHA 54425cb5df86fbb239c4faab624d786f98fb343e
Target service	services/individual-owner-service
Database authority	individual_owner_db
Primary status	IMPLEMENTATION SPECIFICATION – READY FOR DOMAIN BUILD

Purpose

This document is the authoritative backend implementation plan for the Individual Property Owner domain. It translates every approved V5 product area into concrete service ownership, data models, state machines, APIs, events, background workers, cross-service contracts, security controls, testing gates and deployment criteria.

Traceability rule

Product-derived requirements are preserved as product requirements. Engineering additions that are necessary to make the backend safe, operable and certifiable are explicitly identified as engineering additions; they do not silently change the product vision.

0. Document status, authority and reading guide

The V5 product document defines an Individual Property Owner platform for personally owned residential, commercial, land, sale and owner-operated short-let activity. It gives the owner direct control over property records, listings, customers, agreements, payments, maintenance and performance while preserving the owner as the underlying authority.

This backend specification turns that product vision into an implementation contract for the existing CosmicForge Real Estate architecture. It is deliberately more detailed than the product document because backend engineering also requires concurrency rules, idempotency, state transition guards, audit trails, failure recovery, database constraints, service authentication, event contracts, background work and release certification.

Source / baseline	Role in this specification
Individual Property Owner Product Feature Document v5.0	Functional source of truth for owner-facing capabilities, journeys and ecosystem boundaries.
Certified shared Go foundation (F1)	UUIDv7, auth/JWT, service auth, idempotency, outbox, NATS, money, audit, HTTP, DB, telemetry and testing primitives.
Certified API Gateway (F2)	Trusted domain issuer validation, actor propagation, rate limits, public routing and internal-route isolation.
Certified Verification Service (F3)	Identity, property authority and professional/business verification truth.
Certified Media & Document Service (F4)	Signed upload/download, scanning, metadata, access classification and retention.
Certified Notification Service (F5)	FCM push plus provider-neutral email/SMS notification delivery.
Current platform architecture	Seven user-domain services, four shared services, one gateway; isolated databases; REST + NATS.

0.1 Product-derived requirements versus engineering additions

Marker	Meaning
PRODUCT	Directly required by the V5 Individual Property Owner product document.
FOUNDATION	Already implemented/certified by the shared platform foundation.
ENGINEERING ADDITION	Required for correctness, security, reliability, integration or operability but not explicitly stated in the product document.
POLICY CONFIGURATION	Business/legal policy that must be configurable because the product document does not define a universal value.
DEFERRED DEPENDENCY	The domain contract is specified here, but the dependent service/provider must still be implemented.

1. Executive implementation mandate

Build `individual-owner-service` as the sole authoritative domain service for personally owned property operations. The service must support a single owner or personal portfolio, direct operation without compulsory management, optional property-specific delegated management, complete property history and direct customer interaction through the wider Real Estate ecosystem.

- The owner remains the primary authority for personally owned property at all times.
- Property verification and listing publication are separate controlled transitions.
- Every compulsory customer charge must be disclosed before commitment and versioned after acceptance.
- Delegation is property-specific, capability-specific, time-limited, revocable and never grants payout redirection authority by default.
- Financial, tenancy, inspection, maintenance and audit records survive the end of manager, tenant, guest or listing relationships.
- The service owns domain business state but does not own payment ledger truth, verification truth, raw media bytes, system notification delivery or human communications.

1.1 Definition of done for the domain

- Every approved product capability has an owning module, data model and callable API or explicit integration contract.
- Every mutable business lifecycle has a guarded state machine and auditable transition history.

- Every side-effecting retryable command that can duplicate money, publication, booking, signature or external action uses idempotency.
- Every business state change that must propagate to another service writes an outbox event in the same PostgreSQL transaction.
- Marketplace publication is event-driven and never makes Marketplace authoritative for source records.
- All frontend-facing endpoints are represented in OpenAPI and Postman with examples and error cases.
- All financial commands delegate to payment-service; the owner service stores business obligations and projections, never its own independent financial ledger.
- All human-to-human messaging/calls/video use CF Meet contexts; the owner service stores only business context references and authorization.
- All required CI, race, migration, contract, E2E, security and recovery gates pass before promotion to staging.

2. Service architecture and hard boundaries

```
Client apps
   |
   v
API Gateway
   |
   v
individual-owner-service  ---> verification-service
        |                  ---> media-document-service
        |                  ---> notification-service
        |                  ---> payment-service          [deferred full implementation]
        |                  ---> CF Meet                  [messaging / voice / video]
        |                  ---> marketplace-service      [events + authoritative read-back]
        |                  ---> property-manager-service [delegation contracts]
        |                  ---> artisan-service          [maintenance jobs]
        |                  ---> professional-services-service [due diligence]
        |                  ---> admin-service            [moderation / disputes / support]
        |
        +---- individual_owner_db
        +---- outbox -> NATS JetStream -> consumers
```

Concern	Owner / authority	Individual-owner responsibility
Owner account/auth	individual-owner-service	Full authority: account, credentials, sessions, devices, MFA, roles and permissions.
Property record	individual-owner-service	Full authority for personally owned property records and portfolio history.
Identity / property authority verification	verification-service	Creates/links cases and enforces required results; does not duplicate verification truth.
Media/document bytes	media-document-service	Stores `media_id` references and business relationship/access purpose.
Payments/ledger/refunds/payouts	payment-service	Defines business obligations and commands payments; stores payment references/projections only.
System notifications	notification-service	Requests notifications and stores domain notification intent/history references.
Human messaging/calling/video	CF Meet	Creates communication contexts and authorization; does not implement chat/RTC.
Public discovery	marketplace-service	Publishes events/projections; remains source of truth for owner listing.
Delegated management	owner service + property-manager-service	Owner service owns mandate/permission grant; manager service owns manager operational workspace.
Maintenance provider execution	artisan-service	Owner service owns maintenance request and approval; artisan service owns provider execution profile/workflow.
Professional execution	professional-services-service	Owner service owns appointment/mandate relationship; professional service owns provider execution.
Moderation/disputes	admin-service + owning domain	Domain owns case-linked state/evidence; admin service owns platform operator workflow.

2.1 Non-negotiable database rule

`individual-owner-service` receives credentials only for `individual_owner_db`. It must never read or write another service's database and must never depend on a foreign schema. Cross-service state is represented by typed external IDs and service/domain ownership metadata.

2.2 Go module and internal package structure

```
├── integrations/
│   ├── verification/
│   ├── media/
│   ├── notification/
│   ├── payment/
│   ├── cfmeet/
│   ├── marketplace/
│   ├── manager/
│   ├── artisan/
│   ├── professional/
│   ├── maps/
│   ├── esign/
│   ├── external_listings/
│   └── smart_access/
```

3. Domain decomposition

Module	Scope
Account & Owner Profile	Registration, security, identity-verification linkage, owner profile, contacts, accessibility, devices, sessions and account closure.
Banking, Payout & Tax Profile	Verified payout destinations, change controls, property/category payout preferences, tax/profile statement metadata.
Ownership & Authority	Ownership claims, co-owner consent, power-of-attorney/subletting authority, ownership change workflow and authority evidence.
Property & Portfolio	Persistent property/unit/parcel records, physical attributes, assets, history and portfolio summaries.
Property Verification Linkage	Required verification cases, evidence linkage, transaction-ready status and re-verification triggers.
Listings & Publication	Category-specific listing drafts, content, media relationships, moderation state, publication and archive.
Pricing & Fees	Versioned price books, fee lines, refundability, schedules, escalation/discount rules and customer cost snapshots.
Availability & Freshness	Unit/date availability, freshness confirmation, stale-listing pause and reliability signals.
Leads & Marketing	Inquiries, stages, qualification, follow-ups, campaigns, promoted flags and external sync.
Viewing & Access	Slots, requests, attendance, one-time codes, representatives, occupied-property consent and safety reports.
Applications & Screening	Applicant/co-applicant references, consented document sharing, screening criteria/results, decisions, appeals and rental offers.
Agreements & Move-In	Agreement versions, participants, signature process, schedules, inventories, condition evidence and possession.
Tenancy & Rent Operations	Tenancy state, obligations, arrears projection, notices, renewal, move-out and deposit settlement business state.
Utilities & Service Charges	Utility accounts, meter readings, allocation rules, service-charge budgets, bills, evidence and notices.
Commercial Transactions	Commercial-specific use/zoning/facility terms, LOI, offers, fit-out and lease/sale workflow.
Land Transactions	Parcel-specific attributes, lease obligations, professional due diligence, offers, milestones and possession/perfection tracking.
Property Sale & Closing	Buyer offers, conditions, secure data room, due diligence, purchase agreement, milestones and handover.
Short-Let Operations	Calendar inventory, stay pricing, booking operations, guest access, turnover, damage claims and profitability projections.
Manager Delegation	Manager discovery reference, proposals, mandates, permissions, spend limits, reporting, revocation and handover.
Maintenance & Preventive Care	Requests, responsibility, quotes, approvals, provider assignments, work evidence, rework, warranty and asset schedules.
Professional Services	Appointments, mandates, data-room grants, quotations, reports, findings, complaints and ratings.

Module	Scope
Communications Context	CF Meet context authorization/links plus system notification triggers; no local chat storage.
Reports & Market Intelligence	Operational metrics, financial projections, property performance, comparable-market inputs and advisory AI outputs.
Fraud, Privacy & Disputes	Suspicious activity, access restrictions, disputes, evidence, consent, privacy requests and auditability.
Accessibility & Closure	Assisted onboarding, succession/death/incapacity process, transfers, closure and record continuity.
Financing	Partner finance options, owner payout structure preference, consent, financing status references and seller plans.
Insurance	Policy/product references, requirements, claims linkage and insurance evidence.
Smart Access	Access credential requests, device inventory references, grants, expiry, revocation and history.

4. IDs, references, money and time

- All persistent public entity IDs are UUIDv7 generated by this service.
- Cross-domain people/entities carry both owning domain/service and ID; matching email/phone never implies identity equivalence.
- All timestamps are UTC `timestamptz` internally and RFC3339 over APIs/events; property/booking timezone is stored as an IANA name.
- Money is `amount_minor int64 + ISO-4217 currency`; no floating point money.
- Any accepted price/fee snapshot is immutable and linked by version ID to the resulting application/offer/booking/agreement.
- External service references are stored with service name, external ID, lifecycle status projection and last-synced timestamp where needed.

5. Authentication, owner roles and authorization

Because the platform has no central identity service, `individual-owner-service` implements its own account and authentication persistence using the certified shared Go security packages.

Role / principal	Purpose	Core authority
OWNER	Primary individual owner account	Full domain authority subject to verification, co-owner consent and transaction state.
CO_OWNER	Recorded co-owner participant	Consent/vote and access limited to properties and actions explicitly associated with ownership record.
AUTHORIZED_REPRESENTATIVE	POA/estate/temporary representative	Capabilities and expiry derived from authority record; never broader than evidence permits.
WITNESS / SIGNATORY	Agreement/sale signing participant	Document-specific signing access only.
SERVICE_PRINCIPAL	Authenticated platform service	Capability-policy-limited internal API access.
GATEWAY_DELEGATED_ACTOR	End user routed by API gateway	Owner actor context signed by gateway; domain authorization still enforced locally.

5.1 Account/security capabilities

- Email+password, mobile/OTP and supported OAuth/OIDC sign-in adapters.
- Email/phone verification, password reset, recovery codes, trusted devices and active session management.
- Ed25519 access JWT, 15-minute default, rotating hashed refresh tokens, reuse detection and remote revocation.
- TOTP authenticator MFA; SMS/email second factor through Notification Service; step-up auth for payout changes, sensitive documents and property transfers.
- Biometric sign-in is a client capability that unlocks locally protected credentials/session; the backend never receives raw biometric data.
- Suspicious login/device alerts and security event audit.
- Account status: PENDING_VERIFICATION, ACTIVE, RESTRICTED, SUSPENDED, CLOSURE_PENDING, CLOSED.

ENGINEERING ADDITION — authorization decisions

Every command is authorized against owner role, property authority, co-owner consent requirements, manager/representative delegation, transaction state, verification state and any active admin restriction. Authentication alone never grants business authority.

6. `individual_owner_db` logical data model

The schema is organized by bounded context. The table list below is implementation scope, not a suggestion to collapse all concerns into one aggregate. Each table receives created/updated timestamps and appropriate audit/outbox handling.

Area	Table	Purpose
Account & security	owner_accounts	Owner account identity, status, primary contacts, locale, timezone.
Account & security	owner_credentials	Password/OIDC credential metadata; password hash only.
Account & security	owner_sessions	Refresh-token family/session state, device binding, revoked/reuse state.
Account & security	owner_devices	Trusted devices, push endpoint references, risk flags.
Account & security	owner_mfa_methods	TOTP/SMS/email methods and recovery-code metadata.
Account & security	security_events	Login, recovery, session, MFA and suspicious-activity events.
Profile	owner_profiles	Public/private owner profile and preferences.
Profile	owner_contacts	Verified/secondary contacts, emergency/trusted contacts.
Banking	payout_profiles	Payout preference container; no ledger balance.
Banking	payout_accounts	Masked bank metadata + payment-service beneficiary/account reference and verification state.
Banking	payout_change_requests	Step-up, cooling-off and approval workflow for changes.
Banking	tax_profiles	Tax identifiers/profile metadata and reporting preferences.
Ownership	ownership_records	Property authority claim type, holder, percentages and lifecycle.
Ownership	ownership_parties	Owner/co-owner/estate/POA participants and shares.
Ownership	ownership_consents	Action-specific co-owner consent/vote records.
Ownership	authority_documents	Business relationship to media IDs used as evidence.
Ownership	ownership_change_cases	Transfer/succession/change-control process.
Property	properties	Persistent property root record.
Property	property_units	Building/block/floor/unit/room inventory hierarchy.
Property	land_parcels	Plot/parcel/coordinates/boundary-specific data.
Property	property_addresses	Normalized/structured address + visibility level.
Property	property_features	Amenities, utilities, accessibility, facilities and condition attributes.
Property	property_media_links	Relationship between property/listing purpose and media IDs.
Property	property_assets	Appliance/equipment register, warranty/service metadata.
Property	property_history	Business history snapshots/references for major property lifecycle events.
Verification	verification_links	Verification-service case IDs, type, status projection, expiry and subject mapping.
Listing	listings	Listing root by transaction/property type.
Listing	listing_versions	Immutable published/draft content versions.
Listing	listing_media_links	Listing-specific ordered media references.
Listing	listing_publication	Publication/moderation/source projection state.
Pricing	price_schedules	Base price/rate + frequency/currency/time window.
Pricing	fee_definitions	Mandatory/optional fee lines, refundability and payee purpose.
Pricing	pricing_versions	Immutable accepted customer-facing price snapshot.
Pricing	pricing_rules	Escalation, discount, seasonal, promotion and length-of-stay rules.
Availability	availability_state	Unit/property availability and last confirmation.
Availability	availability_calendar	Date-granular short-let / booking inventory.
Availability	freshness_checks	Confirmation reminders, missed checks, automatic pause history.
Leads	inquiries	Customer inquiry references, source, property/listing and stage.
Leads	lead_activities	Notes, follow-ups, qualification answers and stage transitions.
Viewings	viewing_slots	Owner/manager/representative available times.
Viewings	viewing_requests	Requester reference, slot, type, state, consent and instructions.
Viewings	viewing_attendance	Check-in/out, one-time code, no-show and evidence refs.
Applications	rental_applications	Application reference, listing, applicant reference, state and deadlines.
Applications	application_participants	Co-applicant/occupant/roommate/guarantor typed references.
Applications	application_evidence_grants	Purpose/expiry-limited access to external media/doc evidence.
Applications	screening_policies	Versioned owner screening criteria with fairness/human-review controls.
Applications	screening_results	External/internal flags, explanation and review outcome.
Applications	application_decisions	Owner human decision, reasons and appeal state.
Applications	rental_offers	Offer terms, pricing snapshot, conditions and expiry.
Agreements	agreements	Agreement root and business state.
Agreements	agreement_versions	Immutable generated/document versions.
Agreements	agreement_participants	Tenant/co-tenant/guarantor/co-owner/witness participants.
Agreements	signature_requests	E-sign provider refs, signer state, certificate media IDs.
Agreements	move_in_checklists	Possession checklist root.
Agreements	condition_items	Room/item condition and media evidence.

Area	Table	Purpose
Agreements	key_access_inventory	Keys/cards/codes/lockbox/smart-access handover refs.
Tenancy	tenancies	Active tenancy lifecycle and linked agreement/property.
Tenancy	rent_obligations	Business due schedule and payment-service intent/reference projection.
Tenancy	rent_obligation_status_history	Due/paid/partial/overdue/waived/disputed/refunded projection history.
Tenancy	deposit_obligations	Deposit business terms and release conditions.
Tenancy	renewal_cases	Renewal proposal/counterproposal/addendum process.
Tenancy	notices	Notice type, legal template/version, delivery and acknowledgement references.
Tenancy	move_out_cases	Termination/move-out inspection/balance/deposit settlement process.
Utilities	utility_accounts	Property utility provider/meter/responsibility data.
Utilities	meter_readings	Move-in/periodic/move-out readings and media refs.
Utilities	utility_bills	Imported/issued bill references and allocation status.
Utilities	service_charge_budgets	Budget period/categories/expected amount.
Utilities	service_charge_items	Actual/billed/paid evidence and adjustments.
Commercial	commercial_terms	Zoning/use/facility/fit-out/parking/signage etc.
Commercial	commercial_lois	LOI and negotiation lifecycle.
Land	land_terms	Permitted use/lease/restoration/sublease/development obligations.
Transactions	sale_transactions	Property/commercial/land sale lifecycle.
Transactions	sale_offers	Buyer offer/counteroffer/conditions/expiry.
Transactions	data_rooms	Transaction data room root.
Transactions	data_room_grants	Time/purpose-limited grants to professionals/buyers.
Transactions	due_diligence_items	Legal/survey/valuation/inspection findings and resolution.
Transactions	closing_milestones	Payment/consent/perfection/handover checklist.
Short-let	shortlet_settings	Capacity, rules, check-in/out, fees, cancellation, add-ons.
Short-let	shortlet_rates	Date/rate/seasonal pricing rules.
Short-let	shortlet_bookings	Booking business state and payment refs.
Short-let	shortlet_guests	Guest list references and verification projection.
Short-let	turnover_jobs	Cleaning/inspection/supplies/guest-ready workflow.
Short-let	damage_claims	Evidence, claimed amount, response, dispute/refund projection.
Delegation	manager_proposals	Manager proposal/fee/capacity references.
Delegation	manager_assignments	Property-specific manager mandate and lifecycle.
Delegation	manager_permissions	Capability list and spending/emergency limits.
Delegation	manager_activity_refs	Owner-visible manager activity/report references.
Maintenance	maintenance_requests	Property maintenance request and responsibility.
Maintenance	maintenance_quotes	Provider quote refs and comparison.
Maintenance	maintenance_approvals	Owner/manager approval decision and budget.
Maintenance	maintenance_jobs	Job reference to artisan execution + domain state.
Maintenance	maintenance_warranties	Completion/rework/warranty relationship.
Maintenance	preventive_schedules	Recurring asset inspection/service schedule.
Professional	professional_appointments	Professional provider appointment/mandate.
Professional	professional_findings	Signed report references/findings used by owner workflow.
Dispute	domain_disputes	Owner-side dispute case and target entity.
Dispute	dispute_evidence	Media/document references and evidence metadata.
Privacy	consents	Purpose/version/expiry consent records.
Privacy	data_subject_requests	Access/correction/export/deletion/objection workflow.
Reports	report_snapshots	Generated report metadata and downloadable artifact refs.
Reports	market_insights	Comparable inputs, confidence, generated-at and advisory result.
Financing	finance_offers	Partner offer references, consent and owner payout preference.
Insurance	insurance_policies	Policy/product references, requirements and lifecycle.
Insurance	insurance_claims	Claim reference/status/evidence.
Access	access_devices	Smart lock/device/inventory provider refs.
Access	access_grants	Time-limited access purpose/principal/credential status.
Platform	idempotency_records	Shared-go idempotency schema in this database.
Platform	outbox_events	Transactional event outbox.
Platform	processed_events	Consumer de-duplication.
Platform	audit_events	Append-only domain/security audit trail.

6.1 Required database constraints and indexes

- Unique active account per normalized verified primary email/phone within this domain, with soft-closure semantics defined explicitly.
- Unique property ownership root per persistent property record; duplicate claims are represented as conflict cases, not duplicate authoritative property rows.
- Only one active published listing per property/unit + transaction type unless product explicitly allows parallel listing variants.
- Pricing version referenced by accepted application/offer/booking/agreement is immutable.

- Manager assignment intervals and permissions are property-scoped; assignment revocation does not delete historical actions.
- State transitions use compare-and-swap/version columns or row locks to prevent concurrent contradictory decisions.
- External callback/event IDs are unique where consumed; duplicate delivery must be harmless.
- Use partial indexes for open/active states and due workers; use GIN/JSONB only where schemaless attributes are genuinely needed.
- Do not store raw card data, full provider secrets, raw KYC documents or chat message content.

7. Canonical domain state machines

Account

```
PENDING_VERIFICATION -> ACTIVE -> RESTRICTED | SUSPENDED | CLOSURE_PENDING -> CLOSED
RESTRICTED/SUSPENDED -> ACTIVE only through authorized remediation/appeal.
```

Ownership record

```
DRAFT -> EVIDENCE_PENDING -> UNDER_VERIFICATION -> VERIFIED | REJECTED | CONFLICT
VERIFIED -> REVERIFICATION_REQUIRED | TRANSFER_PENDING -> VERIFIED/TRANSFERRED.
```

Property

```
DRAFT -> ACTIVE -> RESTRICTED | TRANSFER_PENDING | ARCHIVED
ACTIVE is not equivalent to listing publication.
```

Listing

```
DRAFT -> READY_FOR_REVIEW -> PUBLISHED -> PAUSED | WITHDRAWN | ARCHIVED
PAUSED -> PUBLISHED after freshness/verification checks.
Published listing may be system-paused for stale availability, moderation or authority expiry.
```

Availability

```
AVAILABLE | RESERVED | APPLICATION_UNDER_REVIEW | OFFER_ACCEPTED | OCCUPIED | TEMP_UNAVAILABLE | OFF_MARKET |
SOLD | WITHDRAWN
```

Inquiry/lead

```
NEW -> ACKNOWLEDGED -> QUALIFIED -> VIEWING | APPLICATION | OFFER | BOOKING | CLOSED_WON | CLOSED_LOST
```

Viewing

```
REQUESTED -> CONFIRMED -> CHECKED_IN -> COMPLETED
REQUESTED/CONFIRMED -> RESCHEDULED | CANCELLED | NO_SHOW
```

Application

```
DRAFT/RECEIVED -> SUBMITTED -> UNDER_REVIEW -> INFO_REQUESTED -> UNDER_REVIEW -> APPROVED | CONDITIONAL |
DECLINED | WAITLISTED | WITHDRAWN | EXPIRED
DECLINED may enter APPEAL/HUMAN_REVIEW where applicable.
```

Rental offer

```
DRAFT -> SENT -> ACCEPTED | DECLINED | EXPIRED | WITHDRAWN
ACCEPTED freezes pricing snapshot and proceeds to agreement.
```

Agreement

```
DRAFT -> REVIEW -> SIGNING -> FULLY_SIGNED -> EFFECTIVE
Any pre-effective state may become CANCELLED; versions immutable after signature invitation.
```

Tenancy

```
PENDING_MOVE_IN -> ACTIVE -> RENEWAL_PENDING | TERMINATION_PENDING -> MOVE_OUT_PENDING -> ENDED
ACTIVE may carry DISPUTE flags without collapsing lifecycle.
```

Rent obligation

```
SCHEDULED -> DUE -> PARTIAL | PAID | OVERDUE | DISPUTED | WAIVED | REFUND_PENDING -> REFUNDED
Payment truth comes from payment-service events/read-back.
```

Deposit

```
REQUIRED -> FUNDED -> HELD -> RELEASE_PENDING | DEDUCTION_PROPOSED -> PARTIAL_RELEASED | RELEASED | DISPUTED
```

Notice

```
DRAFT -> ISSUED -> DELIVERED -> ACKNOWLEDGED | DELIVERY_FAILED | DISPUTED
Legal effective date is policy-driven and not inferred solely from app acknowledgement.
```

Sale transaction

```
INQUIRY -> OFFERING -> OFFER_ACCEPTED -> DUE_DILIGENCE -> CONTRACTING -> CONDITIONS_PENDING -> CLOSING ->
COMPLETED
Any pre-completion state may become CANCELLED/FAILED/DISPUTED subject to terms.
```

Short-let booking

```
INQUIRY/HOLD -> PENDING_APPROVAL | PENDING_PAYMENT -> CONFIRMED -> CHECKED_IN -> CHECKED_OUT -> COMPLETED
May transition to CANCELLED | NO_SHOW | DISPUTED.
```

Manager assignment

```
PROPOSED -> ACCEPTED -> ACTIVE -> SUSPENDED | EXPIRING -> REVOKED | EXPIRED | TERMINATED
Permissions versioned; payout destination change excluded unless separately owner-authorized and step-up
verified.
```

Maintenance request

```
OPEN -> TRIAGED -> QUOTING -> APPROVAL_PENDING -> APPROVED -> ASSIGNED -> IN_PROGRESS -> COMPLETION_REVIEW ->
COMPLETED
May enter REWORK | CANCELLED | DISPUTED.
```

Professional appointment

```
REQUESTED -> QUOTING -> APPOINTED -> IN_PROGRESS -> REPORT_SUBMITTED -> ACCEPTED | DISPUTED | CANCELLED
```

Data subject request

```
SUBMITTED -> VERIFIED_REQUESTER -> PROCESSING -> PARTIALLY_FULFILLED | FULFILLED | DENIED_WITH_REASON ->
CLOSED
```

Account closure

```
REQUESTED -> BLOCKERS_CHECK -> CLOSURE_PENDING -> CLOSED
Open obligations/disputes/required records remain retained and accessible as policy permits.
```

Smart access grant

```
REQUESTED -> APPROVED -> ISSUED -> ACTIVE -> EXPIRED | REVOKED | COMPROMISED
```

8. Public API surface (`/api/v1/owner/*`)

Exact path naming may be normalized during implementation, but the endpoint groups below are the required public contract. All frontend-facing operations must be represented in OpenAPI and Postman and must use the platform error envelope.

Authentication & account

- POST /api/v1/owner/auth/register
- POST /api/v1/owner/auth/login
- POST /api/v1/owner/auth/refresh
- POST /api/v1/owner/auth/logout
- POST /api/v1/owner/auth/forgot-password
- POST /api/v1/owner/auth/reset-password
- POST /api/v1/owner/auth/mfa/enrol
- POST /api/v1/owner/auth/mfa/verify
- GET /api/v1/owner/sessions
- DELETE /api/v1/owner/sessions/{session_id}
- GET/PATCH /api/v1/owner/profile
- POST /api/v1/owner/account/closure

Banking & tax

- GET/POST /api/v1/owner/payout-accounts
- POST /api/v1/owner/payout-accounts/{id}/verify
- POST /api/v1/owner/payout-accounts/{id}/make-default
- DELETE /api/v1/owner/payout-accounts/{id}
- GET/PUT /api/v1/owner/payout-preferences
- GET/PUT /api/v1/owner/tax-profile
- GET /api/v1/owner/statements

Ownership & properties

- GET/POST /api/v1/owner/properties
- GET/PATCH /api/v1/owner/properties/{property_id}
- POST /api/v1/owner/properties/{property_id}/claim
- GET/POST /api/v1/owner/properties/{property_id}/ownership
- POST /api/v1/owner/properties/{property_id}/co-owners
- POST /api/v1/owner/properties/{property_id}/consents
- POST /api/v1/owner/properties/{property_id}/verification
- GET /api/v1/owner/properties/{property_id}/history
- GET/POST /api/v1/owner/properties/{property_id}/units
- GET/POST /api/v1/owner/properties/{property_id}/assets

Listings & pricing

- GET/POST /api/v1/owner/listings

- GET/PATCH /api/v1/owner/listings/{listing_id}
- POST /api/v1/owner/listings/{listing_id}/preview
- POST /api/v1/owner/listings/{listing_id}/publish
- POST /api/v1/owner/listings/{listing_id}/pause
- POST /api/v1/owner/listings/{listing_id}/withdraw
- POST /api/v1/owner/listings/{listing_id}/copy
- GET/PUT /api/v1/owner/listings/{listing_id}/pricing
- GET /api/v1/owner/listings/{listing_id}/price-history
- GET/PUT /api/v1/owner/listings/{listing_id}/availability
- POST /api/v1/owner/listings/{listing_id}/confirm-availability

Leads & viewings

- GET /api/v1/owner/inquiries
- GET/PATCH /api/v1/owner/inquiries/{inquiry_id}
- POST /api/v1/owner/inquiries/{inquiry_id}/qualify
- GET/POST /api/v1/owner/viewing-slots
- GET/PATCH /api/v1/owner/viewings/{viewing_id}
- POST /api/v1/owner/viewings/{viewing_id}/approve
- POST /api/v1/owner/viewings/{viewing_id}/reschedule
- POST /api/v1/owner/viewings/{viewing_id}/cancel
- POST /api/v1/owner/viewings/{viewing_id}/representative

Applications & offers

- GET /api/v1/owner/applications
- GET /api/v1/owner/applications/{application_id}
- POST /api/v1/owner/applications/{application_id}/request-info
- POST /api/v1/owner/applications/{application_id}/decision
- POST /api/v1/owner/applications/{application_id}/offer
- GET/PATCH /api/v1/owner/offers/{offer_id}
- POST /api/v1/owner/offers/{offer_id}/withdraw

Agreements & tenancy

- GET/POST /api/v1/owner/agreements
- GET/PATCH /api/v1/owner/agreements/{agreement_id}
- POST /api/v1/owner/agreements/{agreement_id}/invite-signers
- POST /api/v1/owner/agreements/{agreement_id}/send-for-signature
- GET /api/v1/owner/tenancies
- GET /api/v1/owner/tenancies/{tenancy_id}
- POST /api/v1/owner/tenancies/{tenancy_id}/renewal
- POST /api/v1/owner/tenancies/{tenancy_id}/notices
- POST /api/v1/owner/tenancies/{tenancy_id}/move-out
- GET /api/v1/owner/tenancies/{tenancy_id}/rent-obligations
- GET /api/v1/owner/tenancies/{tenancy_id}/deposit

Utilities & service charge

- GET/POST /api/v1/owner/properties/{property_id}/utilities
- POST /api/v1/owner/properties/{property_id}/meter-readings
- GET/POST /api/v1/owner/properties/{property_id}/service-charge-budgets
- POST /api/v1/owner/service-charge-budgets/{budget_id}/items

Commercial / land / sale

- GET/PUT /api/v1/owner/properties/{property_id}/commercial-terms
- GET/POST /api/v1/owner/transactions/sales
- GET /api/v1/owner/transactions/sales/{sale_id}
- POST /api/v1/owner/transactions/sales/{sale_id}/offers/{offer_id}/decision
- POST /api/v1/owner/transactions/sales/{sale_id}/data-room
- POST /api/v1/owner/transactions/sales/{sale_id}/milestones
- GET/PUT /api/v1/owner/properties/{property_id}/land-terms

Short-let

- GET/PUT /api/v1/owner/properties/{property_id}/shortlet/settings
- GET/PUT /api/v1/owner/properties/{property_id}/shortlet/calendar
- GET/PUT /api/v1/owner/properties/{property_id}/shortlet/rates
- GET /api/v1/owner/shortlet/bookings
- GET/PATCH /api/v1/owner/shortlet/bookings/{booking_id}
- POST /api/v1/owner/shortlet/bookings/{booking_id}/damage-claim
- POST /api/v1/owner/shortlet/bookings/{booking_id}/turnover

Delegation

- GET /api/v1/owner/managers/search
- POST /api/v1/owner/manager-proposals
- GET/POST /api/v1/owner/manager-assignments
- GET/PATCH /api/v1/owner/manager-assignments/{assignment_id}
- POST /api/v1/owner/manager-assignments/{assignment_id}/revoke
- GET /api/v1/owner/manager-assignments/{assignment_id}/activity

Maintenance & professionals

- GET/POST /api/v1/owner/maintenance
- GET/PATCH /api/v1/owner/maintenance/{request_id}
- POST /api/v1/owner/maintenance/{request_id}/request-quotes
- POST /api/v1/owner/maintenance/{request_id}/approve
- GET/POST /api/v1/owner/professional-appointments
- GET/PATCH /api/v1/owner/professional-appointments/{appointment_id}

Reports, disputes, privacy

- GET /api/v1/owner/reports/portfolio

- POST /api/v1/owner/reports/export
- GET /api/v1/owner/market-insights
- GET/POST /api/v1/owner/disputes
- GET /api/v1/owner/disputes/{dispute_id}
- GET/POST /api/v1/owner/privacy/requests
- GET /api/v1/owner/audit/activity

Financing, insurance, smart access

- GET /api/v1/owner/financing/options
- POST /api/v1/owner/financing/consents
- GET /api/v1/owner/insurance/options
- GET/POST /api/v1/owner/insurance/policies
- GET/POST /api/v1/owner/insurance/claims
- GET/POST /api/v1/owner/access/grants
- POST /api/v1/owner/access/grants/{grant_id}/revoke
- GET /api/v1/owner/access/history

8.1 API command rules

- All create/publish/approve/withdraw/decision/financial/external-side-effect commands use `Idempotency-Key` where duplicate execution could be harmful.
- PATCH requests use optimistic concurrency (`version` or `If-Match`) for business objects with concurrent owner/manager/admin modification risk.
- Sensitive actions require step-up authentication freshness; token age alone is not sufficient for payout/property-transfer actions.
- Pagination defaults to cursor-based for unbounded collections.
- Responses expose explicit DTOs; database rows and provider responses are never returned directly.

9. Internal service API contracts

Caller/Dependency	Contract	Purpose
marketplace-service	GET /internal/v1/owner/listings/{id}	Authoritative listing read-back before high-value actions; public projection recovery.
marketplace-service	GET /internal/v1/owner/properties/{id}/public-summary	Fresh public property/authority summary.
verification-service	POST/GET verification APIs	Create/read IDENTITY and PROPERTY_AUTHORITY cases using existing verification-service contract.
media-document-service	Upload/download/status APIs	Media/document lifecycle; owner service stores media IDs only.
notification-service	Send/recipient APIs	System notifications, push endpoints and preferences.
payment-service	Payment intent / obligation / refund / payout APIs	Financial execution and ledger truth. Contract must be implemented before financial modules can certify.
property-manager-service	Manager proposal/assignment/activity APIs	Discovery and operational manager workspace integration.
artisan-service	Maintenance provider quote/job APIs	Provider execution and provider status.
professional-services-service	Professional search/quote/appointment/report APIs	Due-diligence and professional work execution.
admin-service	Moderation/dispute/restriction APIs	Operator actions on owner-domain entities through service API, never DB access.
CF Meet	Context / participants / messaging / call APIs	Human communication context for inquiry, viewing, tenancy, maintenance and transaction.

10. Event catalogue and Marketplace projections

Event	Producer	Consumers	Meaning
owner.account.activated.v1	individual-owner-service	Notification/Admin analytics	Owner account ready.
owner.account.restricted.v1	individual-owner-service	Gateway/Admin/Notification as needed	Account restriction projection.
property.created.v1	individual-owner-service	Marketplace/Admin	Persistent property created; not necessarily publishable.
property.updated.v1	individual-owner-service	Marketplace/Admin	Publicly relevant property fields changed.
property.verification_changed.v1	individual-owner-service	Marketplace	Derived public authority/verification summary changed.
listing.published.v1	individual-owner-service	Marketplace	Create/update search projection.
listing.updated.v1	individual-owner-service	Marketplace	Refresh projection fields.
listing.paused.v1	individual-owner-service	Marketplace	Hide/suspend projection.
listing.withdrawn.v1	individual-owner-service	Marketplace	Remove active discovery projection.
listing.availability_confirmed.v1	individual-owner-service	Marketplace	Update freshness/availability.
viewing.updated.v1	individual-owner-service	Marketplace/Notification	Customer-visible appointment state.
application.updated.v1	individual-owner-service	Marketplace/Notification	Customer-facing application state.
rental_offer.updated.v1	individual-owner-service	Marketplace/Notification	Offer state and deadlines.
tenancy.started.v1	individual-owner-service	Marketplace/Payment/Notification	Occupancy begins.
tenancy.ended.v1	individual-owner-service	Marketplace/Payment/Notification	Occupancy ends.
manager.assignment.created.v1	individual-owner-service	property-manager-service	Owner mandate becomes available to manager service.
manager.assignment.updated.v1	individual-owner-service	property-manager-service	Permissions/limits changed.
manager.assignment.revoked.v1	individual-owner-service	property-manager-service	Immediate access revocation.
maintenance.request.created.v1	individual-owner-service	artisan-service	Approved/eligible work request discovery where applicable.
sale.transaction.updated.v1	individual-owner-service	Marketplace/Notification/Admin	Customer-visible sale state.
shortlet.availability.updated.v1	individual-owner-service	Marketplace	Date inventory projection.
shortlet.booking.updated.v1	individual-owner-service	Marketplace/Notification/Payment	Booking state.
access.grant.revoked.v1	individual-owner-service	Smart-access adapter/Notification	Credential revocation propagation.

10.1 Events consumed

Consumed event family	Owner-service action
verification.*	Update verification-link projections, unlock/lock publication or re-verification-dependent actions.
media.ready/rejected/deleted	Update media usability; automatically prevent publication if required media becomes unavailable.
payment.* / payout.* / refund.*	Update business payment projections, receipts, obligations, deposits, booking/rent/sale status. Never synthesize financial truth.
notification.*	Optional delivery history projection for owner-facing notification history.
manager.*	Manager acceptance/activity/report status where the manager service is authoritative for execution.
maintenance provider/job events	Update quote/job/completion projections.
professional appointment/report events	Update appointment/report projections.
marketplace report events	Stale/duplicate/misleading listing reports routed back to authoritative owner service.

11. Background workers and scheduled jobs

Worker	Responsibility
Outbox relay	Publish committed domain events to NATS with leasing/retry.
Availability freshness	Find due confirmations; send reminders; auto-pause after policy threshold.
Verification expiry	Detect linked verification expiry/reverification requirement and restrict dependent capabilities.
Payout-change cooling-off	Activate verified payout account only after step-up + cooling-off policy passes.
Listing expiry/quality	Pause stale/invalid listings; reconcile missing required media/verification.
Viewing reminders	Schedule reminders and no-show follow-up notifications.
Application deadlines	Expire applications/offers/information requests when deadline passes.
Agreement/signature reconciliation	Read back external e-sign state; never repeat ambiguous signature side effects blindly.
Rent schedule projection	Materialize upcoming obligations from tenancy schedule and request payment intents when due according to policy.
Arrears/reminder	Trigger domain notices/notifications from payment truth and policy.
Renewal/notices	Generate reminders for upcoming expiry/renewal windows.
Move-out/deposit deadlines	Track inspection, response and refund/deduction deadlines.

Worker	Responsibility
Short-let calendar	Expire holds, reconcile external calendar sync, release inventory.
Turnover scheduler	Create cleaning/inspection jobs after checkout or before check-in.
Manager mandate expiry	Revoke expired permissions and notify manager/owner.
Preventive maintenance	Create due preventive tasks from asset schedules.
Privacy retention	Execute retention/deletion/anonymization jobs subject to legal/business hold flags.
Report generation	Produce monthly/quarterly/annual/custom report snapshots.
Smart access expiry	Revoke credentials when business relationship ends.

12. Cross-service and external integration contracts

Integration	Contract / boundary
Verification Service	Use existing `IDENTITY` and `PROPERTY_AUTHORITY` cases. Individual owner service creates cases, attaches media evidence by ID, and enforces verified/expired/rejected outcomes.
Media & Document Service	All documents/media use signed upload flow. Property/listing/application/agreement/data-room relationships store `media_id`; access grants are purpose/time scoped.
Notification Service	FCM selected for push. Email/SMS remain provider-neutral. Owner service requests templates/categories and never calls Twilio/SMTP directly.
Payment Service	Only payment-service may call Paystack or other money providers. Owner service submits business amount/context and receives payment/refund/payout/ledger references and events.
Marketplace Service	Consumes owner listing/property/availability events into discovery projections. Marketplace may call owner service for authoritative read-back.
Property Manager Service	Manager search/proposal data comes from manager domain. Owner service owns the mandate/grant; manager service enforces it in operational workflows.
Artisan Service	Maintenance provider discovery/quotes/execution live in artisan domain; owner service owns property request, responsibility, approval and property history.
Professional Services	Professional discovery/credential/status and execution live in professional domain; owner service owns appointment purpose, transaction link and use of findings.
Admin Service	Admin moderates/reviews through internal APIs with admin actor delegation; admin cannot directly mutate owner DB.
CF Meet	Inquiry/viewing/application/tenancy/maintenance/transaction contexts use CF Meet for chat, voice and video. Store `cfmeet_context_id` and permission state only.
Maps	Provider-neutral geocoding, reverse-geocoding, coordinates validation, directions/area metadata. Persist normalized result + provider/source freshness.
Electronic signature	Provider-neutral adapter inside owner service for signature requests/callback verification/read-back; signed artifacts stored via media service.
External marketplace/calendar sync	Provider-neutral adapters; outbound sync uses idempotency and source-of-truth rules; inbound calendars cannot overwrite authoritative business state without validation.
Smart access	Provider-neutral adapter; device-specific credentials never become permanent platform authorization. Owner-service access grants remain authority.
Financing / insurance partners	Consent-gated partner adapters; owner service stores business references/projections and does not expose unnecessary tenant/buyer financial data.

13. Product-section implementation map (V5 sections 1–30)

The following subsections are the implementation contract for every product area in the Individual Property Owner V5 document. Each subsection states the backend realization and additions required for safe execution.

13.1 Owner Role and Dashboard

PRODUCT requirements

- Portfolio summary across verification, listings, inquiries, viewings, applications, offers, leases, bookings, payments, maintenance, managers, professionals, disputes and reports.
- Calendar and quick actions scoped to owner authority.

Backend implementation

- Dashboard query service backed by read models/materialized projections in individual_owner_db.
- Calendar items derived from authoritative domain tables; no separate calendar authority.

- Quick actions execute normal domain commands with full authorization/idempotency.

Primary tables	Primary public API groups
dashboard projections, calendar items, saved dashboard preferences	GET /api/v1/owner/dashboard, GET /api/v1/owner/calendar

13.2 Registration, Security and Personal Verification

PRODUCT requirements

- Identity checks, recovery, multiple sign-in methods, device/session controls, MFA, re-verification, appeals, verification badges, account status and data rights.

Backend implementation

- Implement domain-owned auth using certified shared Go packages.
- Create verification-service IDENTITY cases; account activation/payout-sensitive actions depend on policy.
- OAuth providers behind adapter; biometric remains client-side authentication unlock.
- Step-up auth freshness for bank changes/document access/property transfer.
- Human-review/appeal records link to verification/admin workflows.

Primary tables	Primary public API groups
owner_accounts, owner_credentials, owner_sessions, owner_devices, owner_mfa_methods, verification_links, security_events	/auth/*, /sessions, /profile, /account/status, /verification

13.3 Owner Banking, Payout and Tax Profile

PRODUCT requirements

- Multiple verified payout accounts, account-name matching, stronger approval/cooling-off for changes, payout preferences, tax/reporting records.

Backend implementation

- Payout account verification/beneficiary truth must come through payment-service/provider adapter, not local trust.
- Store masked account metadata and payment-service beneficiary refs only.
- Cooling-off worker; step-up MFA; owner notifications for changes.
- Tax calculations/rules are configurable; do not hard-code tax law into core schema.

Primary tables	Primary public API groups
payout_profiles, payout_accounts, payout_change_requests, tax_profiles	/payout-accounts, /payout-preferences, /tax-profile, /statements

13.4 Ownership and Authority Records

PRODUCT requirements

- Sole/joint/family/inherited/probate/POA/subletting authority, co-owner percentages/consents, documents, beneficial owner/representative, disputes/reverification, ownership transfer.

Backend implementation

- Model ownership parties and evidence as first-class records.
- Action-specific co-owner consent policy determines whether publication/price/sale/manager actions may proceed.
- Verification-service owns PROPERTY_AUTHORITY decision; owner service owns claim/workflow.
- Ownership transfer is a guarded case, never direct owner_id update.

Primary tables	Primary public API groups
ownership_records, ownership_parties, ownership_consents, authority_documents, ownership_change_cases	/properties/{id}/ownership, /co-owners, /consents, /ownership-transfer

13.5 Permanent Property and Portfolio Records

PRODUCT requirements

- Persistent property record, address/coordinates/unit/parcel refs, categories/features, media, asset register, complete history, bulk upload.

Backend implementation

- Persistent property ID survives listing/tenant/manager changes.
- Separate property root, units and land parcels.
- Bulk import is asynchronous with validation/error report and idempotent row matching.
- Media bytes never stored locally; property media links reference media-service IDs.

Primary tables	Primary public API groups
properties, property_units, land_parcels, property_addresses, property_features, property_assets, property_media_links, property_history	/properties, /properties/{id}/units, /properties/{id}/assets, /properties/imports

13.6 Property Verification and Listing Authenticity

PRODUCT requirements

- Document/inspection/registry/survey/payout-beneficiary checks, verification levels/expiry, duplicate/conflicting claims, geotag/live media, public summary.

Backend implementation

- Orchestrate required verification types by property/category/transaction policy.
- Duplicate property/media/conflicting owner claims produce conflict cases requiring admin/verification workflow.
- Geotag/freshness metadata captured with media relation; do not treat client metadata as trusted without validation.
- Public verification summary is a projection with limitations/expiry.

Primary tables	Primary public API groups
verification_links, property_history, domain_disputes	/properties/{id}/verification, /properties/{id}/verification-summary

13.7 Listing Creation for Every Property Category

PRODUCT requirements

- Residential/commercial/land/sale/short-let listings, property vs transaction type, fields/media/inventory/pricing, draft-preview-publish-pause-withdraw-copy-archive, campaigns/external sharing.

Backend implementation

- One listing aggregate with category-specific typed detail tables/JSON-schema-validated extensions.
- Publish command validates authority, verification, required media, pricing disclosure and availability.
- Published immutable version emitted to Marketplace; edits create new version.
- External sync is asynchronous and must not become source of truth.

Primary tables	Primary public API groups
listings, listing_versions, listing_media_links, listing_publication	/listings, /listings/{id}/preview, /publish, /pause, /withdraw, /copy

13.8 Complete Price and Fee Configuration

PRODUCT requirements

- Base price, mandatory charges, refundability, payment cycles, escalation/discount/seasonal rules, all-in preview, history, accepted-cost freeze.

Backend implementation

- Price/fee model is versioned and immutable once accepted.
- All-in cost calculator is deterministic and shared between preview and acceptance.
- Money uses integer minor units + currency.
- Unknown/legal fee categories require configured policy; hidden fees are rejected.

Primary tables	Primary public API groups
price_schedules, fee_definitions, pricing_versions, pricing_rules	/listings/{id}/pricing, /listings/{id}/price-history, /listings/{id}/cost-preview

13.9 Availability and Listing Freshness

PRODUCT requirements

- Availability states, scheduled confirmations, auto-pause, report handling, unit/room/date inventory, quality measures.

Backend implementation

- Freshness policy configurable by listing/category.
- Workers send reminders then system-pause stale listing with reason.
- Short-let uses date calendar; long-term uses state + effective dates.
- Marketplace receives availability/freshness events.

Primary tables	Primary public API groups
availability_state, availability_calendar, freshness_checks	/listings/{id}/availability, /confirm-availability

13.10 Marketing, Inquiries and Lead Management

PRODUCT requirements

- Inquiries, acknowledgement, qualification, pipeline, reminders/notes/conversation history, matching, promoted listings, external sync.

Backend implementation

- Inquiry business state stored locally; actual human message content in CF Meet.
- Automated acknowledgements use Notification Service or CF Meet depending on message type.
- Promoted listing flag references Marketplace/commercial capability; payment, if any, must use payment-service.
- Lead qualification fields must be lawful/configurable, not unrestricted owner-defined discrimination.

Primary tables	Primary public API groups
inquiries, lead_activities	/inquiries, /inquiries/{id}/qualify, /campaigns

13.11 Viewing and Access Management

PRODUCT requirements

- Viewing slots, owner/manager/representative, virtual/live/physical/open-house, codes, attendance/no-show, occupied privacy, feedback/safety, remote inspection.

Backend implementation

- Viewing aggregate links to CF Meet for live video.
- One-time access credential issued through smart-access adapter or server-generated code with TTL.
- Occupied-property consent gate before access instructions.
- Representative authority is checked against assignment/authorization record.

Primary tables	Primary public API groups
viewing_slots, viewing_requests, viewing_attendance, access_grants	/viewing-slots, /viewings/{id}/*

13.12 Residential Applications and Tenant Selection

PRODUCT requirements

- Applications/participants, minimum necessary data, evidence request, screening criteria/alternative evidence, human decision, rental offer, protected history.

Backend implementation

- Applicant source record may originate Marketplace; owner service stores domain application and typed applicant refs.
- Purpose/time-limited evidence grants via media service.
- Screening policy version captured at decision time; automated flags never become sole final decision.
- Decision reasons, appeal/human review and retention rules audited.

Primary tables	Primary public API groups
rental_applications, application_participants, application_evidence_grants, screening_policies, screening_results, application_decisions, rental_offers	/applications, /applications/{id}/request-info, /decision, /offer

13.13 Agreement and Move-In

PRODUCT requirements

- Agreement template/terms, payment schedule/rules/inventory/condition/deposit, multi-party signing, e-sign, version history/certificate, move-in evidence/meters/keys.

Backend implementation

- Agreement templates are jurisdiction/policy-versioned; legal content managed as controlled configuration, not hard-coded strings.
- E-sign provider adapter with verified callbacks and read-back for ambiguous outcomes.
- Signed document/certificate stored in media service.
- Move-in condition evidence is immutable once both parties acknowledge, with dispute addendum instead of overwrite.

Primary tables	Primary public API groups
agreements, agreement_versions, agreement_participants, signature_requests, move_in_checklists, condition_items, key_access_inventory	/agreements, /agreements/{id}/send-for-signature, /move-in

13.14 Rent, Deposit and Financial Management

PRODUCT requirements

- Supported payments, schedules, rent ledger states, late fees/reminders, deposit ledger/release, payouts/reconciliation/chargebacks/refunds/statements, reports.

Backend implementation

- Owner service owns rent/deposit business obligations only; payment-service owns transaction/ledger truth.
- Every money movement initiated through payment-service idempotently.
- Unknown provider outcomes are read back before retry.
- Late fees and deposit release rules are policy-configured and auditable.
- Financial projections reconcile to payment events/read-back.

Primary tables	Primary public API groups
rent_obligations, deposit_obligations, payout_profiles, report_snapshots	/tenancies/{id}/rent-obligations, /deposit, /statements

13.15 Renewal, Rent Review, Notices and Move-Out

PRODUCT requirements

- Renewal proposal/counter, rent change explanation/market context, signed addendum, notices, move-out, damage/deposit deductions/refund, return-to-market.

Backend implementation

- Renewal/counter is a separate versioned case; accepted terms generate addendum/agreement version.
- Notice templates and effective dates are policy-controlled; delivery and acknowledgement recorded separately.
- Deposit deduction requires itemized evidence; dispute freezes disputed settlement path, not necessarily unrelated funds.
- Listing/property returns AVAILABLE only after move-out completion and owner confirmation.

Primary tables	Primary public API groups
renewal_cases, notices, move_out_cases, condition_items, deposit_obligations	/tenancies/{id}/renewal, /notices, /move-out

13.16 Utilities and Service-Charge Management

PRODUCT requirements

- Utility providers/meters/responsibility, bills/receipts, shared-meter allocation/challenge, service-charge budget/actuals, announcements/alerts.

Backend implementation

- Utility/service-charge records are domain accounting projections, not payment ledger.
- Meter readings may require media evidence and author attribution.
- Allocation rules are versioned; disputes preserve original calculation.
- Announcements use Notification Service; human replies use CF Meet.

Primary tables	Primary public API groups
utility_accounts, meter_readings, utility_bills, service_charge_budgets, service_charge_items	/properties/{id}/utilities, /meter-readings, /service-charge-budgets

13.17 Commercial Property Rent and Sale

PRODUCT requirements

- Commercial use/zoning/facility data, corporate applicant verification, LOI/offer/counter, fit-out/rent-free/escalation/break/service-charge/insurance/subletting/repairs, lease/sale closing.

Backend implementation

- Commercial applicant business verification remains external/domain cross-service reference.
- Commercial terms captured in typed structured terms and immutable offer/contract versions.
- Fit-out/obligation calendar is auditable and can generate maintenance/professional tasks.
- Commercial sale reuses sale transaction engine with commercial extensions.

Primary tables	Primary public API groups
commercial_terms, commercial_lois, sale_transactions, agreements	/commercial-terms, /commercial-lois, /transactions/sales

13.18 Land Rent, Lease and Sale

PRODUCT requirements

- Parcel/title/survey/boundary/access/utilities/use, lease terms/obligations, professional appointments, offers, title search/charting/encumbrance, milestones/possession/perfection.

Backend implementation

- Land parcel root separated from generic unit inventory.
- Survey/title evidence stored in media service and professional findings linked.
- Perfection/registration is tracked as milestone status; backend must not claim governmental completion without evidence/provider status.

Primary tables	Primary public API groups
land_parcels, land_terms, professional_appointments, sale_transactions, closing_milestones	/properties/{id}/land-terms, /transactions/sales

13.19 Property Sale, Offers and Closing

PRODUCT requirements

- Buyer inquiry/viewing/proof of funds/offer, counter/conditions, reservation/deposit/history, secure data room, due diligence, purchase agreement, milestones/final statement/handover/perfection.

Backend implementation

- Sale aggregate is separate from listing and survives listing withdrawal.
- Data room uses purpose/time-limited media grants and access audit.
- Buyer proof-of-funds/verification references are not copied unnecessarily into owner DB.
- Reservation/milestone money uses payment-service.
- Closing completion requires configured checklist and authorized evidence.

Primary tables	Primary public API groups
sale_transactions, sale_offers, data_rooms, data_room_grants, due_diligence_items, closing_milestones	/transactions/sales, /data-room, /milestones

13.20 Personally Operated Short-Let

PRODUCT requirements

- Calendar, rates, capacity/rules/fees/cancellation/add-ons, inquiries/verification/payment/access/comms/extension, turnover, damage/deposit dispute/refund, reports, external calendar sync.

Backend implementation

- Date inventory supports hold/confirm race prevention with transactional locking.
- Booking pricing snapshot immutable after confirmation except explicit modification workflow.
- Guest human communications via CF Meet; system notifications via notification-service.
- External calendar sync is best-effort adapter and cannot override paid/confirmed authoritative booking without conflict resolution.

Primary tables	Primary public API groups
shortlet_settings, shortlet_rates, availability_calendar, shortlet_bookings, shortlet_guests, turnover_jobs, damage_claims	/shortlet/settings, /calendar, /rates, /bookings

13.21 Property Manager Marketplace and Delegation

PRODUCT requirements

- Search/compare/proposals, assign properties/managers, capability permissions, spend/emergency limits, role types, activity/performance/expenses/complaints, revoke/handover.

Backend implementation

- Manager marketplace/search owned by property-manager-service; owner service stores proposal/assignment refs and authoritative owner grant.
- Capability matrix is explicit; no generic 'manager can do everything'.

- Revocation must immediately block internal API capability checks and emit event.
- Manager cannot redirect owner funds; payout changes remain owner step-up flow.

Primary tables	Primary public API groups
manager_proposals, manager_assignments, manager_permissions, manager_activity_refs	/managers/search, /manager-assignments

13.22 Maintenance, Artisans and Preventive Care

PRODUCT requirements

- Requests/priority/responsibility, quotes, approval/budget/delegation, provider attendance/progress/evidence/invoice/completion/rework/warranty, pay provider, preventive schedules.

Backend implementation

- Owner service owns property request/responsibility/approval; artisan service owns provider execution.
- Payment-service pays provider only against authorized job/amount.
- Work completion can be challenged/reworked without overwriting evidence.
- Preventive scheduler ties tasks to property assets.

Primary tables	Primary public API groups
maintenance_requests, maintenance_quotes, maintenance_approvals, maintenance_jobs, maintenance_warranties, preventive_schedules	/maintenance, /maintenance/{id}/request-quotes, /approve

13.23 Professional Services and Due Diligence

PRODUCT requirements

- Find/compare/appoint professionals, mandate, data-room access, pay fee, track report/licence/conflict/quote/status/signed report, use findings, ratings/complaints.

Backend implementation

- Professional service owns professional profile/credential execution.
- Owner service stores appointment purpose and findings references.
- Media grants are least-privilege and expire.
- Professional fee payment through payment-service.

Primary tables	Primary public API groups
professional_appointments, professional_findings, data_room_grants	/professional-appointments

13.24 Communication and Notifications

PRODUCT requirements

- Automated notifications, in-app messaging context, profile/contact preferences, SMS/email/WhatsApp, voice/video/masked contact.

Backend implementation

- Notification Service handles system push/email/SMS.
- CF Meet handles human messaging/voice/video and privacy relay.
- WhatsApp is provider capability to be added behind notification/communication policy when selected; do not hard-code now.
- Owner DB stores communication context IDs and preference/domain metadata, not chat content.

Primary tables	Primary public API groups
owner_profiles, owner_contacts	Notification public preference endpoints plus CF Meet context APIs

13.25 Owner Finance, Reports and Market Intelligence

PRODUCT requirements

- Income/expenses/net/arrears/occupancy/yield, short-let ADR/profitability, price recommendations/comparables/trends, reports, AI vacancy/renewal/maintenance insights, tax/statutory calendar.

Backend implementation

- Reports combine owner-domain business state with payment-service financial projections using API/event data, never cross-DB queries.
- AI/market advice must store model/version/input provenance/confidence and be advisory only.
- Do not automate tenant selection or material owner decisions solely from AI.
- Large reports generated asynchronously and stored via media service.

Primary tables	Primary public API groups
report_snapshots, market_insights	/reports/*, /market-insights

13.26 Fraud, Privacy, Safety and Disputes

PRODUCT requirements

- Unauthorized manager/forged docs/false ownership/duplicate property/payment diversion/abuse, occupied-property privacy.

Backend implementation

- Domain dispute cases link to admin operations workflow and evidence.
- Sensitive data purpose/access controls enforced through media grants and route authorization.
- Fraud signals may restrict actions but final significant decisions require review path.
- Every privileged read and restriction transition audited.

Primary tables	Primary public API groups
domain_disputes, dispute_evidence, consents, data_subject_requests, audit_events	/disputes, /privacy/requests

13.27 Accessibility, Assisted Use and Account Closure

PRODUCT requirements

- Concierge onboarding/photography/scanning/remote help, death/incapacity/estate successor, transfer/role removal/closure and record continuity.

Backend implementation

- Assisted actions are performed under attributed support/admin actor context, never shared generic accounts.
- Succession uses ownership-change case and verification evidence.
- Closure checks blockers, revokes auth, preserves legally/business-required records and produces export where permitted.

Primary tables	Primary public API groups
ownership_change_cases, data_subject_requests, owner_accounts	/account/closure, /ownership-transfer

13.28 Rent Financing, Mortgage and Payment-Plan Options

PRODUCT requirements

- Partner rent instalment/advance, owner payout structure, finance-supported tenant status with privacy, mortgage buyer conditions, seller plans, repayment/settlement/dispute, total cost disclosure.

Backend implementation

- Financing is partner-integrated and consent-gated.

- Owner sees only minimum necessary status; sensitive applicant finance details remain with partner/customer domain.
- Seller payment plan is versioned contractual schedule and payment-service executes money.
- Total cost/conditions must be present in listing/offer pricing snapshots where advertised.

Primary tables	Primary public API groups
finance_offers, pricing_versions, closing_milestones	/financing/options, /financing/consents

13.29 Insurance and Property Protection

PRODUCT requirements

- Insurance products/requirements, policy comparison/renewal/storage, claims/evidence/report/status/payout/complaint, workmanship/fidelity cover visibility.

Backend implementation

- Provider-neutral insurance adapter/partner contract.
- Insurance product/policy truth remains partner-referenced; owner service stores policy/claim refs and business requirements.
- Premium/claim payouts use payment-service only when platform handles money.

Primary tables	Primary public API groups
insurance_policies, insurance_claims	/insurance/options, /policies, /claims

13.30 Smart Access and Remote Property Controls

PRODUCT requirements

- Time-limited access codes for viewing/tenant/guest/manager/provider, key/card/lock inventory, access history/alerts, remote approval, automatic end.

Backend implementation

- Access grant authority remains owner service; device adapter only executes credential provisioning/revocation.
- Grant expiry is tied to business relation and worker-enforced.
- Compromised credential revokes immediately and generates security notification/audit.
- No remote access persists beyond lawful business period.

Primary tables	Primary public API groups
access_devices, access_grants, key_access_inventory	/access/grants, /access/history

14. Primary end-to-end journeys

14.1 Onboard and verify a property

1. Owner registers/authenticates in individual-owner-service.
2. Owner completes account verification; service creates IDENTITY case in verification-service.
3. Owner creates persistent property record and ownership claim.
4. Owner uploads authority documents/media through media-document-service signed upload flow.
5. Owner service creates PROPERTY_AUTHORITY verification case and attaches evidence refs.
6. Verification result event/read-back updates verification link projection.
7. When policy requirements pass, property becomes transaction-ready; this does not automatically publish a listing.
8. Owner creates listing, pricing and availability, previews full customer cost and publishes.
9. Publication transaction writes listing state + outbox event atomically; Marketplace consumes `listing.published.v1`.

14.2 Rent a property directly

1. Published listing is discovered through Marketplace projection.

2. Customer inquiry/viewing/application reaches owner domain with typed marketplace actor/customer references.

3. Owner communicates through CF Meet and system notifications are sent through Notification Service.

4. Owner reviews purpose-limited application evidence and records human screening decision.

5. Owner creates rental offer with immutable pricing snapshot.

6. Accepted offer creates agreement; signatures occur through e-sign adapter; artifacts stored via media service.

7. Rent/deposit obligations are created in owner domain and payment intents are executed by payment-service.

8. After required initial conditions are confirmed, tenancy enters PENDING_MOVE_IN then ACTIVE.

9. Move-in checklist, meters, keys/access and condition evidence are recorded.

10. During tenancy, payment-service events update obligations; owner can manage maintenance, notices and renewal.

11. Move-out/deposit settlement completes; property becomes available only after owner inspection/confirmation.

14.3 Sell or lease land / commercial property

1. Owner maintains verified parcel/commercial details and publishes correct transaction listing.

2. Buyer/business inquiry and offer are received; required verification/proof status is referenced, not copied unnecessarily.

3. Owner appoints professionals; data-room grants provide least-privilege document access.

4. Due-diligence findings and conditions are tracked against sale/lease transaction.

5. Payment milestones are requested through payment-service; unknown outcomes are reconciled.

6. Closing/possession/perfection checklist is completed with evidence; transaction completes only when configured conditions pass.

7. Marketplace availability/listing projection is withdrawn/sold accordingly.

14.4 Operate a short-let

1. Owner configures calendar inventory, rates, stay rules, fees and cancellation terms.

2. Marketplace exposes date inventory projection.

3. Booking hold uses transaction/locking to prevent overbooking; confirmation requires required guest verification/payment state.

4. CF Meet handles guest conversation; Notification Service handles system reminders.

5. Smart-access credential is issued for the stay period only.

6. Turnover job is scheduled around check-out/check-in.

7. Damage claim workflow may use deposit/payment-service and disputes without overwriting evidence.

8. Booking completes, access expires, reports update occupancy/revenue projections.

14.5 Appoint and revoke a property manager

1. Owner searches manager profiles from property-manager-service and reviews proposal.

2. Owner creates property-specific assignment with capabilities, spend limit, emergency rule, term and report schedule.

3. Owner service emits assignment-created event; manager service activates operational workspace.

4. Every manager action hitting owner service is authorized against active assignment and capability.

5. Permission changes are versioned and immediately propagated.

6. Revocation terminates capabilities and access grants immediately; property history remains intact.

7. Handover/report data is retained and ownership authority never changes.

15. Engineering additions required for completeness

Why this section exists

The V5 product document describes the desired product experience. The controls below are implementation necessities that make those features safe and production-operable. They are additions to the backend specification, not changes to the approved product vision.

Engineering addition	Reason
Optimistic concurrency / fencing	Prevent two owner/manager/admin requests from making contradictory decisions on listing, pricing, application, assignment or transaction state.
Idempotency on commands	Required for publication, decisions, signature requests, external sync, payments, bookings, access grants and provider actions.
Transactional outbox	Every cross-service event tied to a business state change is persisted atomically with that state.
Processed-event de-duplication	All consumed NATS events are at-least-once and must be harmless when duplicated.
Unknown outcome handling	External payment/e-sign/smart-access/calendar operations use read-back before repeating ambiguous side effects.
Capability-based service auth	Internal callers receive only the operations needed; valid service JWT does not equal unrestricted access.
Policy engine/configuration	Legal/business windows, fee rules, verification requirements, consent thresholds, retention and notice templates are configuration/versioned data, not constants.
Audit immutability	Security, authority, consent, price, decision, payout, access, manager and dispute changes are append-only attributable audit events.
Retention/legal hold	Deletion/anonymization jobs respect open dispute, transaction, financial and required record holds.
Privacy purpose binding	Document access is linked to active purpose, transaction and expiry, with revocation.
Rate limits / abuse controls	Gateway and service controls for auth, uploads, search-like actions, OTP/recovery and sensitive commands.
Feature flags	Provider/partner features such as WhatsApp, financing, insurance, external listing sync and smart access can be disabled safely per environment.
Schema/backfill rollout	Use expand-migrate-contract pattern for backwards-compatible production migrations.
Business clock / timezone	Property/booking local time kept explicitly to prevent due-date/check-in errors.
Data-quality reconciliation	Workers detect orphaned external refs, stale projections, failed event delivery and status mismatches.
Manual review / admin escalation	Significant fraud/verification/privacy/screening conflicts have an operator path rather than silent automatic final decisions.

16. Security, privacy and data protection implementation

Control	Implementation
Credential protection	Argon2id passwords; refresh tokens hashed; Ed25519 JWTs; secret values never logged.
Step-up actions	Payout-account changes, property ownership transfer, sensitive document grants, high-risk access and selected money commands require recent MFA.
Least privilege	Owner/co-owner/representative/manager/service/admin access evaluated per property, capability and business state.
Sensitive documents	Media service access classification + short-lived signed URLs + audit; owner service stores only media refs.
Purpose limitation	Application, buyer, tenant, guest and professional evidence access has purpose, grantee, expiry and revocation.
PII minimization	Do not copy external verification/provider data unless needed for domain function; store normalized result/reference.
Audit	Append-only actor, action, target, before/after summary, request/trace IDs; secrets and raw documents excluded.
Fraud	Duplicate property/media/ownership conflicts, payment-diversion indicators and unauthorized manager claims route to restrictions/manual review.
Account recovery	Recovery events invalidate risky sessions as policy dictates; remote logout supported.
Closure	Authentication revoked but required historical business records retained; export/deletion requests tracked independently.

17. Financial architecture and Payment Service dependency

The product requires rent, deposits, short-let payments, sale milestones, professional/maintenance fees, refunds, payouts, chargebacks and reconciliation. These capabilities cannot be certified end-to-end until `payment-service` is implemented. `individual-owner-service` must nevertheless be built against a stable payment contract from the start.

Owner domain owns	Payment service owns
Business obligation: what is due, why, property/tenancy/booking/sale context, configured schedule.	Payment intents, provider execution, ledger/journals, transaction truth.
Deposit release conditions and owner/tenant dispute state.	Actual held/received/refunded money state and settlement.
Payout preference selection by property/category.	Beneficiary validation, payout execution, reconciliation and unknown-outcome recovery.
Customer-facing price snapshot and agreement amounts.	Collection references, receipts/transaction IDs and financial audit.
Refund business approval/amount/reason.	Refund provider execution and final money state.

DEFERRED DEPENDENCY

Financial product sections may be implemented behind interfaces/fakes in development, but they cannot receive production PASS until payment-service and the relevant regulated/payment provider arrangements are implemented and certified.

18. Reliability, observability and recovery

- Expose `/health`, `/ready`, `/version` for API and worker processes.
- Use structured `slog` JSON with service/environment/request_id/trace_id and secret redaction.
- OpenTelemetry spans around HTTP, DB transactions, NATS publish/consume and external provider calls.
- Explicit outbound timeouts; no external network call while holding unnecessary DB locks.
- Workers use leases, `FOR UPDATE SKIP LOCKED`, expired-lease recovery and bounded retry.
- Readiness checks only hard dependencies required to serve traffic; optional providers may degrade by feature flag.
- Dead-letter/manual-review strategy for poison events/jobs rather than infinite retry.
- Reconciliation metrics for Marketplace projection lag, payment projection mismatch, external calendar conflict and manager/consumer event lag.

Metric family	Examples
HTTP	request rate, latency, 4xx/5xx, auth failures, rate-limit events
DB	pool saturation, tx retries, lock wait, query latency
Events	outbox backlog/age, publish failure, consumer lag, duplicate count
Workers	claim latency, retry count, lease recovery, final failure
Business	published listings, stale pauses, application decision time, occupancy, overdue obligations, booking conflicts
Security	MFA failures, suspicious login, payout-change attempts, denied document grants, manager authorization denials

19. Configuration and secrets

Variable family	Purpose
APP_ENV / LOG_LEVEL / HTTP_ADDR	Runtime environment and API bind.
DATABASE_URL[_FILE] / DB_MAX_CONNS	`individual_owner_db` only.
REDIS_*	Rate/ephemeral coordination where service-level use is needed.
NATS_URL / NATS_*	Event streams/subjects and outbox relay.
SERVICE_AUTH_*	Service identity and trusted peer public keys.
OWNER_TOKEN_ISSUER / OWNER_TOKEN_AUDIENCE / OWNER_AUTH_PRIVATE_KEY[_FILE]	Domain user JWT issuer/signing.
VERIFICATION_SERVICE_URL	Internal verification client.
MEDIA_SERVICE_URL	Internal media/document client.
NOTIFICATION_SERVICE_URL	Internal notification client.
PAYMENT_SERVICE_URL	Internal payment client; may be disabled in early dev.
CFMEET_*	CF Meet base URL/service credentials/config gate.

Variable family	Purpose
MARKETPLACE_SERVICE_URL	Authoritative read-back/admin operations if synchronous call required.
MANAGER_SERVICE_URL / ARTISAN_SERVICE_URL / PROFESSIONAL_SERVICE_URL	Cross-domain dependency URLs.
MAPS_PROVIDER / MAPS_*	Provider-neutral map/geocode configuration.
ESIGN_PROVIDER / ESIGN_*	Electronic signature adapter.
SMART_ACCESS_PROVIDER / SMART_ACCESS_*	Smart lock/access adapter.
EXTERNAL_LISTING_PROVIDERS	Opt-in external listing/calendar providers.
POLICY_CONFIG_VERSION	Business/legal policy pack version.
OTEL_*	Telemetry exporter/sampling.

20. Migrations, Docker, deployment and branch flow

- Own Goose migrations under `services/individual-owner-service/migrations`.
- Use sqlc + explicit SQL; generated code is service-local.
- Provide `/app/api`, `/app/worker`, `/app/migrate` in a distroless non-root production image.
- Docker is for CI/deployment; local development does not require Docker.
- Update staging/production Compose with digest-pinned individual-owner-service image only after image workflow is green.
- GHCR image tags may exist, but deployment consumes immutable digest.
- Branch flow remains `feature/* -> development -> staging -> main`.
- Development -> staging requires affected CI gates green. Staging -> main requires deployed staging migrations, readiness, E2E and rollback/recovery proof.

21. OpenAPI and Postman frontend handoff

Postman remains the official human-facing frontend/QA handoff. OpenAPI remains the machine-readable contract. Swagger UI is not required.

- Populate `02 Individual Owner` in `postman/CosmicForge_RealEstate.postman_collection.json`.
- Group requests by Authentication, Properties, Listings, Pricing, Availability, Leads, Viewings, Applications, Agreements, Tenancy, Commercial/Land/Sales, Short-let, Delegation, Maintenance, Professionals, Reports/Privacy and Smart Access.
- Every frontend route must have method/path, description, auth, headers, request body, success example, relevant error examples and tests.
- Create requests should automatically capture IDs (`property_id`, `listing_id`, `application_id`, `tenancy_id`, etc.) into environment variables.
- Internal/admin/provider webhook routes live under clearly separated `_Internal`, `_Admin`, `_Webhooks` folders and are not presented as frontend endpoints.
- CI checks route parity between implemented handlers, OpenAPI operations and Postman requests.

22. Test and certification matrix

Gate	Required proof
Auth	registration/login/refresh/logout; wrong issuer/audience/signature; session reuse; MFA; remote revoke; cross-domain token misuse.
Authorization	owner/co-owner/representative/manager/service/admin capability matrix; revoked/expired assignment; property ownership transfer.
Migrations	empty -> current; down/up where safe; constraints/indexes; sqlc diff clean.
Property/ownership	duplicate claim, co-owner consent race, verification expiry, transfer workflow, bulk import idempotency.
Listing	publish prerequisites, concurrent publish/update, stale auto-pause, withdrawal, Marketplace event contract.
Pricing	all-in calculation, immutable accepted snapshot, mixed currency rejection, fee history.
Availability	unit/date conflict, short-let double-book prevention, freshness policy.
Viewings	slot collision, occupied consent, access code expiry, no-show.
Applications	evidence grants, screening human decision, appeal, retention.

Gate	Required proof
Agreement	version/signature concurrency, provider callback duplicate, unknown outcome read-back, signed artifact linkage.
Financial projections	payment event duplication/out-of-order, unknown state, refund/payout projection reconciliation; no duplicate owner-side commands.
Tenancy	renewal/counter race, notices, move-out/deposit dispute, return-to-market gate.
Commercial/Land/Sale	offer/counter state, data-room access, due diligence, closing milestones.
Short-let	hold/booking race, extension, cancellation, external calendar conflict, turnover, damage claim.
Delegation	permission limits, spend threshold, emergency rule, revocation immediate effect, manager action attribution.
Maintenance	quote/approval/rework/warranty; artisan event duplication; payment authorization.
Professionals	mandate, data-room grants, report status, revoked access.
Privacy	purpose/expiry grants, data request flow, closure holds, privileged access audit.
Smart access	grant/expiry/revoke/compromised credential and adapter outage.
Events/outbox	broker outage, crash after DB commit, crash after publish, duplicate event, poison event.
E2E	onboard+verify+publish; rent direct; sale/land; short-let; manager assignment/revocation.
Security	gitleaks, govulncheck, Trivy image/IaC, secrets absent from logs, rate-limit abuse.

22.1 Service PASS criteria

- All unit/integration/contract/race tests pass with zero required skips.
- PostgreSQL integration/migration tests run against real PostgreSQL in CI.
- NATS recovery/de-dup tests pass.
- Postman/OpenAPI route drift check passes.
- Docker image builds, runs non-root, smoke tests, and has zero unresolved HIGH/CRITICAL fixed vulnerabilities according to project policy.
- Critical E2E journeys pass through real service processes and certified shared services.
- No direct cross-service database access or direct imports of another service's internal packages.
- Repository clean and branch heads verified before promotion.

23. Implementation workstreams and closure order

Workstream	Scope
IO-1 Service bootstrapping & local auth	Go module, config, migrations framework, API/worker/migrate binaries, local account/auth/session/MFA/device/security.
IO-2 Owner profile, banking & tax	Profile/contact/preferences, payout refs/change controls, tax metadata, notification hooks.
IO-3 Ownership & property core	Ownership parties/consents/change case, property/unit/parcel/assets/media relations, bulk import.
IO-4 Verification, listings, pricing & availability	Verification orchestration, listing versions/publication, cost engine, availability/freshness, Marketplace events.
IO-5 Leads, viewing & communication contexts	Inquiry pipeline, viewing schedule/attendance/access, CF Meet contexts, system notifications.
IO-6 Applications, screening, offers	Application participants/evidence grants, criteria, decisions, appeal and offers.
IO-7 Agreements, signatures & move-in	Agreement versions, e-sign adapter, condition evidence, meter/key/access handover.
IO-8 Tenancy, rent/deposit, renewal & move-out	Business obligations, payment-service contracts/projections, notices, renewals, deposit settlement.
IO-9 Utilities & service charges	Meters, bills, allocation, budgets, announcements.
IO-10 Commercial, land & sale	Typed terms, LOI/offers, data room, due diligence, closing/perfection.
IO-11 Short-let	Calendar/rates/booking/guest/turnover/damage/external calendar.
IO-12 Manager delegation	Proposals/assignments/capabilities/limits/revocation/property-manager integration.
IO-13 Maintenance & professionals	Work request/approval, artisan/professional integrations, warranty/findings.
IO-14 Reports, disputes, privacy & closure	Reporting, AI advisory provenance, disputes, DSR, succession/closure.
IO-15 Financing, insurance & smart access	Partner refs/consent, insurance claims, smart access grants/adapters.
IO-16 Full certification	All contracts/Postman/events, migration/race/E2E/security/image/CI gates; staging promotion only after PASS.

23.1 Workstream closure format

```
WORKSTREAM:
STATUS: PASS | PARTIAL | FAIL
COMMIT:
IMPLEMENTED:
DATABASE / MIGRATIONS:
PUBLIC APIs:
INTERNAL APIs:
EVENTS PUBLISHED / CONSUMED:
WORKERS:
POSTMAN / OPENAPI:
TESTS RUN:
TEST RESULT:
KNOWN LIMITATIONS:
NEXT WORKSTREAM:
```

24. Final acceptance checklist

☐ Account/auth is domain-owned and no identity-service is introduced.

☐ Owner remains authoritative for personally owned properties; manager/representative capabilities never silently replace ownership.

☐ Every V5 section 1–30 has implementation coverage.

☐ All five primary user journeys are executable end-to-end.

☐ All property/listing/customer/tenancy/transaction histories remain attributable and durable.

☐ Verification, media, notification, payment and human communications respect service boundaries.

☐ Marketplace receives projections through contracts/events; it never becomes source of truth.

☐ Payment provider access is only through payment-service.

☐ CF Meet is the only human messaging/voice/video authority.

☐ Every sensitive document grant is purpose/time scoped and auditable.

☐ Every external side-effect path is idempotent or has explicit unknown-outcome recovery.

☐ Every business state machine rejects illegal transitions under concurrency.

☐ Postman provides complete frontend handoff for every public endpoint.

☐ CI includes lint, static analysis, race tests, migrations, sqlc drift, contracts, security, images and E2E.

☐ Staging promotion uses certified commit/image digests; main remains production-approved only.

Appendix A. Product source traceability

Product source	Product area	Backend implementation location
Section 1	Owner Role and Dashboard	§13.1 in this specification
Section 2	Registration, Security and Personal Verification	§13.2 in this specification
Section 3	Owner Banking, Payout and Tax Profile	§13.3 in this specification
Section 4	Ownership and Authority Records	§13.4 in this specification
Section 5	Permanent Property and Portfolio Records	§13.5 in this specification
Section 6	Property Verification and Listing Authenticity	§13.6 in this specification
Section 7	Listing Creation for Every Property Category	§13.7 in this specification
Section 8	Complete Price and Fee Configuration	§13.8 in this specification
Section 9	Availability and Listing Freshness	§13.9 in this specification
Section 10	Marketing, Inquiries and Lead Management	§13.10 in this specification
Section 11	Viewing and Access Management	§13.11 in this specification
Section 12	Residential Applications and Tenant Selection	§13.12 in this specification
Section 13	Agreement and Move-In	§13.13 in this specification
Section 14	Rent, Deposit and Financial Management	§13.14 in this specification
Section 15	Renewal, Rent Review, Notices and Move-Out	§13.15 in this specification

Product source	Product area	Backend implementation location
Section 16	Utilities and Service-Charge Management	§13.16 in this specification
Section 17	Commercial Property Rent and Sale	§13.17 in this specification
Section 18	Land Rent, Lease and Sale	§13.18 in this specification
Section 19	Property Sale, Offers and Closing	§13.19 in this specification
Section 20	Personally Operated Short-Let	§13.20 in this specification
Section 21	Property Manager Marketplace and Delegation	§13.21 in this specification
Section 22	Maintenance, Artisans and Preventive Care	§13.22 in this specification
Section 23	Professional Services and Due Diligence	§13.23 in this specification
Section 24	Communication and Notifications	§13.24 in this specification
Section 25	Owner Finance, Reports and Market Intelligence	§13.25 in this specification
Section 26	Fraud, Privacy, Safety and Disputes	§13.26 in this specification
Section 27	Accessibility, Assisted Use and Account Closure	§13.27 in this specification
Section 28	Rent Financing, Mortgage and Payment-Plan Options	§13.28 in this specification
Section 29	Insurance and Property Protection	§13.29 in this specification
Section 30	Smart Access and Remote Property Controls	§13.30 in this specification
Primary journeys	Onboard/verify; rent direct; sell/lease land & commercial; operate short-let; appoint manager	§14
Ecosystem connections	Marketplace, managers, artisans, professionals, platform operations	§2, §10, §12

Appendix B. Decisions intentionally left configurable

- Exact legal agreement/notice text and jurisdiction-specific deadlines.
- Exact co-owner consent thresholds by ownership type/action.
- Late-fee percentages, grace periods and arrears escalation policy.
- Deposit handling/release timing and any regulated safeguarding arrangements.
- Verification provider/vendor and registry availability.
- Production email/SMS/WhatsApp provider selection.
- Electronic-signature provider selection.
- Smart-lock/access provider selection.
- External listing/calendar sync providers.
- Financing and insurance partner/product availability.
- Tax rates/statutory rules and any filing automation.
- AI model/provider and recommendation thresholds.

These items are deliberately configurable or deferred because the product document defines the capability but does not provide one universal business/legal value. Implementing hard-coded assumptions would make the backend less correct, not more complete.

Appendix C. Backend implementation verdict

IMPLEMENTATION BASELINE

The certified foundation (F1–F5) is complete and available to this domain. The Individual Property Owner service itself is currently scaffold-only and should be built using workstreams IO-1 through IO-16. This specification is intended to be sufficient for Codex/backend engineers to implement the domain without inventing architecture or violating the already-certified platform boundaries.

Appendix D. Product-to-current-platform gaps and required closure

This appendix is mandatory. These are approved product requirements that are not fully satisfied by the current certified foundation or by the domain-local identity decision. They must be closed, explicitly changed by product decision, or carried as a known launch limitation; they must not be silently treated as implemented.

Gap / requirement	Current position	Required closure	Status
Cross-domain single identity / multi-role account	Product says one verified personal identity can hold multiple approved platform roles. Current architecture deliberately keeps accounts domain-local with no central identity service.	Within this service, support OWNER/CO_OWNER/REPRESENTATIVE roles. For cross-domain identity, either accept separate accounts as a deliberate product deviation or add a future explicit account-linking/SSO ADR without recreating shared DB coupling.	OPEN PRODUCT/ARCHITECTURE DECISION
Business context in owner onboarding	Product allows business registration/TIN/director/beneficial-owner checks when the user acts for an organisation. Individual-owner service is for personally owned property; business-owned property belongs to business-operations-service.	Allow verified organisation/representative context only where it supports an individual owner authority case; route business-owned property operations to business-operations-service.	DESIGN RULE REQUIRED
Production identity/KYC/liveness provider	Product requires NIN/passport/licence checks, selfie liveness and face-to-document matching. Verification Service currently ships sandbox provider only in non-production.	Select and implement production verification adapter(s) behind existing provider interface; certify callbacks/read-back, privacy, failure and manual review.	BLOCKS PRODUCTION VERIFICATION
Cross-domain duplicate identity detection	Product calls for duplicate-identity checks, but domain-local accounts intentionally do not share identity tables.	Perform duplicate checks within Individual Owner domain now. Any cross-domain duplicate detection requires privacy-preserving explicit account-linking/risk contract or future architecture decision.	OPEN PLATFORM CAPABILITY
Notification opened / acknowledged receipts	Product requires notification history for delivery, opening, acknowledgement and failed delivery. F5 currently records provider acceptance/failure; delivery/read receipt ingestion is not implemented.	Extend notification-service receipt ingestion and public/internal contracts, or explicitly narrow launch requirement. Owner service should only project statuses notification-service can truthfully provide.	OPEN FOUNDATION ENHANCEMENT
WhatsApp notifications	Product requests consent-based WhatsApp notifications. No WhatsApp provider is selected/implemented.	Add provider-neutral WhatsApp channel/capability with consent, template, secure-link, retry and opt-out policy when a provider is selected.	DEFERRED PROVIDER
Off-platform payment / unapproved bank-detail warnings	Product requires automatic warning when a user requests off-platform payment or sends unapproved bank details. Human messages live in CF Meet.	CF Meet must expose message-safety/risk events or an approved moderation hook; owner service consumes risk events and can restrict/alert without storing chat content.	CF MEET DEPENDENCY
Payment, deposit, refund, payout, bank-name match	Product financial journeys require provider execution and payout verification. payment-service is not implemented yet.	Implement and certify payment-service before production closure of rent/deposit/sale/short-let/provider-payment flows.	BLOCKS FINANCIAL PASS
Production email/SMS	Product requires email/SMS notifications. F5 is provider-neutral; production providers are not selected and default to disabled.	Select Nigerian-compatible production providers, implement/configure adapters, certify delivery/failure/retry and secrets.	DEFERRED PROVIDERS
OAuth providers	Product names Google, Apple, Facebook and LinkedIn sign-in. Shared auth primitives exist but domain provider adapters are not yet implemented.	Implement OIDC/OAuth adapters with account-linking safeguards, verified-email rules, nonce/state/PKCE and recovery behavior.	DOMAIN IMPLEMENTATION

Gap / requirement	Current position	Required closure	Status
Electronic signatures	Product requires electronic signatures and signature certificate. No provider selected.	Implement provider-neutral e-sign adapter with callback verification, read-back on ambiguity, signer identity, version pinning and media artifact storage.	DEFERRED PROVIDER
Smart locks / digital access	Product requires remote/time-limited property access where supported. No device provider selected.	Implement provider-neutral smart-access adapter; owner-service remains grant authority; provider only executes issuance/revocation.	DEFERRED PROVIDER
External listing / social / calendar sync	Product allows approved external marketplace and calendar sync. No providers selected.	Implement adapters as opt-in features with idempotent outbound sync, conflict handling and source-of-truth protection.	DEFERRED PROVIDERS
Financing and insurance partners	Product includes rent finance, mortgage/payment plans and insurance/protection. No partners are selected.	Keep consent/reference contracts in owner domain; integrate providers only after commercial/regulatory partner decisions.	DEFERRED PARTNERS
AI price / vacancy / renewal / maintenance insights	Product includes advisory AI recommendations. No model/provider governance is implemented.	Add model/provider-neutral insight adapter with provenance, model/version, confidence, monitoring, opt-out where applicable and strict non-final-decision controls.	DEFERRED CAPABILITY
Jurisdiction-specific agreements/notices/fees	Product requires state-appropriate agreements/notices and lawful charges but does not define the legal content or deadlines.	Create versioned policy/template packs approved by legal/product; backend enforces selected policy version rather than hard-coding law.	POLICY CONFIGURATION

Appendix D.1 Launch rule for unresolved gaps

- A feature with an unresolved blocking dependency must be feature-flagged or unavailable in production; the API must not falsely report success.
- The UI/Postman/OpenAPI must reflect actual supported behavior, not aspirational provider capability.
- Any deliberate deviation from the V5 product requirement must be recorded as an explicit product/architecture decision before production certification.
- No deferred provider is allowed to leak its vendor-specific types into core domain business logic.

END OF SPECIFICATION